

ADVANCED SQL NULL FUNCTIONS

Q1 Show all employees with their Salary. If salary is NULL, display 0

```
SELECT
    employee-id,
    name,
    IF NULL (salary, 0) AS salary-with-default
FROM Employees;
```

employee-id	name	salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2 Show employees names with their department. If department is NULL, show 'Not Assigned'

```
SELECT
    employee-id,
    name,
    IF NULL (department, 'Not Assigned') AS department-name
FROM Employees;
```

employee-id	name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Q3 Find orders with NULL customer-id using ISNULL()

```
SELECT
  order-id,
  customer-id
FROM orders
WHERE ISNULL(customer-id);
```

order-id	customer-id
103	NULL

Q4 Show all orders. If delivery-date is NULL, show 'Pending'.

```
SELECT
  order-id,
  customer-id,
  IFNULL(delivery-date, 'Pending') AS delivery-status
FROM orders;
```

order-id	customer-id	delivery status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Q5 Show all students and their grades. Replace NULL with 0.

```
SELECT
  student-id,
  name,
  IFNULL(grade, 0) AS final-grade
FROM students;
```

student-id	name	final-grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6 Count Students who havent been graded

```
SELECT  
COUNT (*) AS not-graded-count  
FROM Students  
WHERE grade IS NULL;
```

not-graded-count
1

Q7 Show product name, price and final price after discount

```
SELECT  
product-id,  
name,  
Price - IFNULL(discount, 0) AS final-price  
FROM Products;
```

Product-id	name	final-price
S01	keyboard	25
S02	mouse	10
S03	Monitor	100

Q8 Count how many customers have no email

```
SELECT  
COUNT (*) AS missing-email-count  
FROM Customers  
WHERE email IS NULL;
```

missing-email-count
2

Q9 Show all customers with email. If NULL, display 'No Email'

```
SELECT  
customer - id,  
name ,  
IF NULL (email, 'no Email') AS email - display  
FROM Customers ;
```

Customer - id	name	email - display
1	Linda	No Email
2	Joseph	Joseph @ mail . com
3	Nia	No Email

Q10 Show payment details with method replaced by 'unknown' if NULL

```
SELECT  
payment - id ,  
IF NULL (method, 'unknown') AS method - display ,  
status  
FROM Payments ;
```

Payment - id	method - display	Status
301	Credit	NULL
302	Paypal	Success
303	unknown	failed

Q11 Show items and their quantity. Display 0 if NULL

```
SELECT  
item - id ,  
item - name ,  
IF NULL (quantity, 0) AS quantity checked  
FROM Inventory ;
```

item - id	item - name	quantity - checked
1	PEN	0
2	Notebook	150
3	Eraser	0

Q12 Show employee ID and the first available value among bonus or commission

```
SELECT  
emp-id,  
COALESCE (bonus, commission) AS first-available-reward  
FROM Employee-Extra;
```

emp-id	first-available-reward
1	300
2	100
3	NULL

Q13 Count classes that don't have a room assigned

```
SELECT  
COUNT (*) AS no-room-count  
FROM Classes  
WHERE room IS NULL;
```

no-room-count
2

Q14 Show attendance records with status. Replace NULL with "Not Marked".

```
SELECT  
Student-id,  
date,  
IF NULL (status, 'Not Marked') AS attendance-status  
FROM Attendance
```

Student-id	date	attendance-status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-06-01	Absent

Q15 show account id, account type (or 'unknown'), and balance (or 0)

```
SELECT
  account -id,
  IF NULL (account -type, 'unknown') AS type -display,
  IF NULL (balance, 0) AS balance - checked
FROM Bank -Accounts ;
```

account -id	type -display	balance - checked
A 1	Savings	0
A 2	Current	5000
A 3	Unknown	2000

Q16 show all projects with a start date. If start-date is NULL, display 'TBD'

```
SELECT
  Project -id,
  title,
  IF NULL (start -date, 'TBD') AS start -display
FROM Projects ;
```

Project -id	title	start -display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Q17 Display Reviews showing comment (or 'No comment') and rating (or 0)

```
SELECT
  review -id,
  Product -id
  IF NULL (comment, 'No comment') AS comment -display,
  IF NULL (rating, 0) AS rating - display
FROM Reviews ;
```

Review -id	Product -id	comment -display	rating -display
1	501	Great Product	4
2	502	No comment	0
3	503	Works fine	3

Q18 Show the Supplier contact number using COALESCE

```
SELECT  
Supplier - id,  
name,  
COALESCE (phone, alt phone, 'no contact') AS contact - number  
FROM Suppliers;
```

Supplier - id	name	Contact - number
1	Global Goods	1 234 567 89
2	Best Supplies	9876 54321
3	Value Source	No contact

Q19 Show all users and replace NULL preference with defaults

```
SELECT  
user - id,  
IF NULL (theme, 'Light') AS theme - set,  
IF NULL (language, 'English') AS language - set,  
IF NULL (timezone, 'UTC') AS timezone - set  
FROM User - Settings;
```

User - id	theme - set	language - set	timezone - set
1	light	English	UTC
2	Dark	English	UTC + 1
3	Light	English	UTC

Q20 Show the maintenance log with default values for missing information

```
SELECT  
  record-id,  
  machine-id,  
  IF NULL (issue, 'unknown issue') AS issue-log  
  IF NULL (technician, 'not Assigned') AS technician-name  
FROM Maintenance;
```

record-id	machine-id	issue-log	technician-name
1	m101	Overheating	Not Assigned
2	m102	Unknown Issue	Not Assigned
3	m103	Jammed	Alex